

MediTrack AI

Medicine Reminder & Tracker Project Overview

1. Executive Overview

MediTrack AI is a premium Flutter mobile application developed to help users manage their medicines, schedule medication reminders, track dose history, and monitor medication adherence. It combines Flutter and Dart with Firebase, REST API integration, local storage, state management, and a dedicated notification system.

2. Project Purpose

The project addresses the practical problem of missed or poorly managed medication schedules by providing a centralized platform for medicine information, scheduled reminders, dose recording, and adherence tracking.

3. Core Objectives

- Provide a responsive Flutter mobile experience.
- Allow users to manage medicine schedules.
- Deliver scheduled medicine reminders.
- Maintain medicine and dose-history records.
- Provide health information through REST API integration.
- Demonstrate scalable architecture and reusable UI components.

4. Application Architecture

The application separates presentation, state management, business services, data access, models, routing, and reusable UI components. Riverpod manages state, GoRouter handles navigation and authentication guards, while repositories and services isolate data and platform functionality.

Layer	Responsibility
Screens	Application pages and user interaction flows
Widgets	Reusable UI components
Providers	Application and feature state management using Riverpod
Repositories	Firestore and REST API data access
Services	Authentication and notification services

Layer	Responsibility
Models	Medicine, user, health-tip and dose-record data structures
Routes	GoRouter navigation and authentication guards
Core	Theme, constants and shared configuration

5. Major Application Screens

Screen	Primary Function
Splash	Animated branding and authentication redirect
Login	Email/password and Google authentication
Sign Up	Account creation with validation
Forgot Password	Firebase password-reset workflow
Home	Dashboard with medicines, statistics and health tips
Medicines	Medicine list with pause/resume and deletion
Add/Edit Medicine	Medicine details, dosage, schedule and frequency
History	Calendar-based dose history and adherence charts
Profile	User information, theme settings and sign out
Health Tips	Health tips retrieved through REST API

6. Firebase & Data Management

Firebase Authentication provides email/password authentication, password reset and Google Sign-In. Cloud Firestore stores user profiles and medicine records with real-time data streams.

7. REST API Integration

The application integrates the AdviceSlip REST API to retrieve random health tips. Requests use the HTTP package with timeout handling.

8. Medicine Reminder System

A dedicated notification service schedules timezone-aware daily medicine reminders. Android notification actions include Taken, Snooze 10 minutes and Dismiss. Editing a medicine cancels and reschedules its reminders.

9. Data Models

Entity	Key Information
MedicineModel	ID, name, dosage, type, frequency, reminder times, meal preference, start/end dates, notification status, pause status, weekly schedule, notes and dose history.
DoseRecord	ID, scheduled time, taken time and dose status.
UserModel	User ID, name, email and profile photo.
HealthTipModel	Health-tip information retrieved through the integrated REST API.

10. UI/UX & Design System

The interface follows Material 3 with light, dark and system themes, glassmorphism containers, premium typography, animated buttons, skeleton loaders and responsive layouts.

11. Technology Stack

Category	Technology
Framework	Flutter
Language	Dart
Design	Material 3
State Management	Riverpod
Navigation	GoRouter
Authentication	Firebase Authentication + Google Sign-In
Cloud Database	Cloud Firestore
REST API	AdviceSlip API
Notifications	Flutter Local Notifications + Timezone
Local Storage	Shared Preferences

12. Future Scope

- AI-assisted medication guidance.
- Prescription scanning using OCR.
- Family-member medication management.
- Cloud push notifications.
- Doctor appointment integration.

13. Project Outcome

MediTrack AI demonstrates a complete Flutter application combining responsive UI, navigation, validation, CRUD operations, local persistence, REST API integration, Firebase services, and notification scheduling for practical medication management.